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REVIEW ARTICLE

Allogenic stem cell transplant-associated acute graft versus host disease: a computational drug discovery text mining approach using oral and gut microbiome signatures

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Abstract

Purpose Acute graft versus host disease (aGVHD) is a major cause of non-relapse morbidity and mortality post-allogenic hematopoietic stem cell transplant (HSCT). Using conventional literature search and computational approaches, our objective was to identify oral and gut bacterial species associated with aGVHD, potentially affecting drug treatment via lipopolysaccharide (LPS) pathways.

Methods Medline, PubMed, PubMed Central, and Google Scholar were searched using MeSH terms. The top 100 hits per database were curated, and 25 research articles were selected to examine oral and gut microbiomes associated with health, HSCT, and aGVHD. Literature search validation, aGVHD drug targets, and microbial metabolic pathway identification were completed using BioReader, MACADAM, KEGG, and STRING programs.

Results Our review determined that (1) oral genera *Rothia*, *Solobacterium*, and *Veillonella* were identified in HSCT patients' stool and associated with aGVHD; (2) shifts in gut *enterococci* profiles were determined in HSCT-associated aGVHD; (3) gut microbiome dysbiosis prior or during HSCT and lower Shannon diversity index at time of HSCT were also associated with increased risk of aGVHD and transplant related death; and (4) Coriobacteriaceae family was negatively correlated with gut aGVHD, whereas *Eubacterium limosum* was associated with decreased risk of chronic GVHD relapse.

Additionally, we identified molecular pathways related to TLR4/ LPS, including candidate aGVHD drug targets, impacted by oral and gut bacterial taxa.

Conclusion Reduced microbial diversity reflects higher severity and mortality rate in HSCT patients with aGVHD. Multi-omics approaches to decipher oral and gut microbiome associations will be critical for developing aGVHD preventive therapies.

Keywords Oral microbiome · Gut microbiome · Hematopoietic stem cell transplant · Acute graft versus host disease · Text mining · Computational drug discovery

Introduction

Acute GVHD overview

Acute graft versus host disease (aGVHD) is the major cause of non-relapse morbidity and mortality after allogenic hematopoietic stem cell transplant (HSCT) [1]. Incidence of aGVHD in HSCT patients ranges from 17 to 31% and depends on risk factors including age, cancer type, immunodeficiency, donor-recipient match, and prophylaxis Wojnar [2].

HSCT is the curative option for most acute leukemias, and since aGVHD occurs frequently during treatment, the development of preventive therapies is paramount. Patients who are refractory to the standard steroid therapy for aGVHD are difficult to treat. Patients with aGVHD typically experience an

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11% higher mortality rate, 16 days longer length of hospital stay, and an average increase of $73,000 in medical costs, regardless of age group, with a 40% 1-year survival rate for those with grades 3–4 aGVHD [3].

Acute GVHD inflammatory process

Targets of aGVHD include mucosal linings in the gut, skin, and liver but also affect eyes, mouth, genitalia, and lungs [4]. Donor T-cells and donor antigen presenting cells (APCs) within the graft can either attack residual cancer cells or healthy host tissues [5]. This process has three phases: (i) afferent phase; (ii) donor T cell activation, differentiation, and migration phase; and (iii) effector phase:

(i) In the afferent phase, damage to host tissue usually occurs due to the conditioning regimen (i.e., chemotherapy and radiotherapy) and the release of proinflammatory cytokines, thereby activating recipient APCs.
(ii) In the second phase, donor-derived T cells are activated by donor and recipient APCs and by inflammatory cytokines after recipient antigens are expressed on host cells. APCs present the antigen to a cluster of differentiation 4 (CD4) positive T-cells. The antigen binds to major histocompatibility (MHC) class 2 molecules. Monocyte derived interleukin-1 (IL-1) stimulates T-helper cells, resulting in the release of IL-2 and interferon-gamma. Increased MHC class 2 molecules occur in epithelial cells, macrophages, dendritic cells activating T-cells, and natural killer cells. IL-2 reacts with MHC class 1 positive targets, thereby activating cytotoxic CD8 positive T-cells.
(iii) In the effector phase, symptoms start to manifest as immune effector cells and cytokines begin to cause loss of self-tolerance [6].

aGVHD refers to the dermatitis, hepatitis, and enteritis occurring within 100 days post-HSCT, while chronic GVHD occurs roughly after 100 days of HSCT [7]. Chronic GVHD can be due to unresolved aGVHD, and there must be at least one associated symptom for diagnosis [8]. Severe aGVHD frequently affects the gut, targeting intestinal stem cells (ISCs) and Paneth cells [9].

Impact of conditioning therapy on incidence and severity of aGHVD

Myeloablative conditioning has been associated with a higher incidence of early, more severe aGVHD, and higher mortality. However, myeloablative conditioning alone does not explain reported high incidences of aGVHD in patients receiving reduced intensity conditioning (RIC) or non-myeloablative conditioning (up to 60% developing grades 2–4 aGVHD) [10], although some studies have reported less severe GVHD and reduced mortality associated with RIC or non-myeloablative conditioning [11].

Total body irradiation (TBI), graft source, and conditioning regimen likely have a combined potential for predicting aGVHD risk [12]. A large retrospective study ($N = 2985$ HSCT patients) reported a complex predictive pattern of pre-transplant comorbidities (e.g., age, organ dysfunctions, obesity, diabetes, inflammatory bowel disease) that were associated with aGVHD incidence, severity, and mortality [13]. Also, a higher hematopoietic cell transplantation comorbidity index (HCT-CI) was associated with increased risk of grades 3–4 aGVHD [13].

Oral/gut microbiome dysbiosis and mucosal epithelium integrity

Oral mucosa integrity

The nature and structure of a healthy oral microbiome have been previously described [14]. Oral bacterial taxa associated with health are presented in Table 1a. Oral bacterial species can influence the host cell oral epithelium's cellular immune responses, disrupt the biofilm in oral cavity, and evade inflammatory responses [15]. Under inflammatory conditions in oral cavity (e.g., periodontal disease, oral mucositis), species such as *P. gingivalis* and *F. nucleatum* may contribute to dysbiosis of the oral microbiome [15].

Gut mucosa integrity

A healthy gut microbiome has been described elsewhere [14], and bacterial taxa associated with health are presented in Table 1b. Higher normal oral flora to gut flora transfer rates in people with certain diseases have been previously reported, and it has been established that such transfer is important for the immune homeostasis of gut mucosa [14].

In the course of allogenic HSCT conditioning and antibiotic therapy, the gut mucosa can be damaged, and the microbiota can become unbalanced, leading to impaired intestinal microbiota function and reduced bacterial diversity [16]. Damage to the gut mucosa and microbial dysbiosis may prime the immune system for heightened response to graft, leading to severe aGVHD. Such damage results in translocation of bacteria and their byproducts, causing further damage [17].

Both lymphoid (allogeneic donor T-cells and recipient T-reg cells) and myeloid cells (donor and recipient APC cells) participate in gut aGVHD [18]. In aGVHD, Paneth cells and ISCs are damaged by activated donor T cells within days after transplant [19], involving a complex interplay of interactions between gut microbiota, immune system, and gut mucosa [18] (Fig. 1). Short-chain fatty acid butyrate (SCFA-butyrate)

Table 1 Oral and gut microbiome profiles in health

a. Oral microbiome		
Phylum	Family/Genus	PubMed ID
Actinobacteria	*Actinomycetaceae/ Actinomyces*	HMP, 2012ab; 20,003,481; 30,747,106
	Atopobiaceae/ Olsenella & Atopobium	30,747,106
	Bifidobacteriaceae/ Bifidobacterium	30,747,106
	Corynebacteriaceae/ Corynebacterium	HMP, 2012ab; 20,003,481; 30,747,106
	Micrococcaceae/ Rothia	HMP, 2012ab; 20,003,481; 30,747,106
Bacteroidetes	*Flavobacteriaceae/ Capnocytophaga*	HMP, 2012ab; 20,003,481; 30,747,106
	Porphyromonadaceae/ Porphyromonas	HMP, 2012ab; 20,003,481; 30,747,106
	Prevotellaceae/ Prevotella	HMP, 2012ab; 20,003,481; 30,747,106
	Tannerellaceae / Tannerella	30,747,106
Firmicutes	*Aerococcaceae/ Abiotrophia*	30,747,106
	No Family Assigned/ Gemella	30,747,106
	Carnobacteriaceae/ Granulicatella	HMP, 2012ab; 20,003,481; 30,747,106
	Enterococcaceae/ Enterococcus	30,747,106
	Erysipelotrichidae/ Solobacterium	30,747,106
	Eubacteriaceae/ Eubacterium	30,747,106
	Lachnospiraceae/ Catonella & Oribacterium & Shuttleworthia	30,747,106
	Lactobacillaceae/ Lactobacillus	30,747,106
	Peptostreptococcaceae/ Peptostreptococcus	30,747,106
	Peptoniphilaceae/ Parvimonas	30,747,106
	Selenomonadaceae/ Centipeda & Selenomonas	30,747,106
	Streptococcaceae/ Lactococcus & Streptococcus	30,747,106; HMP, 2012ab; 20,003,481
	Veillonellaceae/ Dialister & Megasphaera & Veillonella	30,747,106
Fusobacteria	*Fusobacteriaceae/ Fusobacterium*	HMP, 2012ab; 20,003,481; 30,747,106
	Leptotrichiaceae/ Leptotrichia	30,747,106
Proteobacteria	*Burkholderiaceae/ Lautropia*	30,747,106
	Campylobacteraceae/ Campylobacter	30,747,106
	Cardiobacteriaceae/ Cardiobacterium	30,747,106
	Neisseriaceae/ Eikenella & Kingella & Neisseria	30,747,106; HMP, 2012ab; 20,003,481
	Pasteurellaceae/ Haemophilus & Aggregatibacter	HMP, 2012ab; 20,003,481; 30,747,106
Spirochaetes	*Spirochaetaceae/ Treponema*	20,656,903; 30,747,106
b. Gut microbiome		
Actinobacteria	*Actinomycetaceae/ Actinomyces*	30,747,106
	Bifidobacteriaceae/ Bifidobacterium	30,232,462; 30,747,106
	Coriobacteriaceae/ Collinsella	30,232,462; 30,747,106
	Eggerthellaceae/ Eggerthella	30,747,106
	Micrococcaceae/ Rothia	30,747,106
Bacteroidetes	*Bacteroidaceae/ Bacteroides*	25,517,225; 30,232,462; 30,747,106
	Porphyromonadaceae	30,232,462
	Prevotellaceae/ Prevotella	30,232,462; 30,747,106

Table 1 (continued)

	Rikenellaceae/ Alistipes	30,232,462; 30,747,106
	Selenomonadaceae/ Mitsuokella	30,747,106
Firmicutes	*Akkermansiaceae/ Akkermansia*	30,747,106
	Carnobacteriaceae/ Granulicatella	30,747,106
	Clostridiaceae/ Clostridium	30,747,106; 25,517,225; 26,556,275
	Erysipelotrichidae/ Solobacterium	30,232,462; 30,747,106
	Eubacteriaceae/ Eubacterium	26,556,275; 30,747,106
	Lachnospiraceae/ Anaerostipes & Blautia & Butyrivibrio & Coprococcus & Dorea & Oribacterium & Pseudobutyrivibrio & Roseburia	30,747,106; 26,556,275; 30,232,462
	Lactobacillaceae	25,517,225
	Desulfovibrionaceae/ Bilophila & Desulfovibrio	30,747,106
	Enterobacteriaceae/ Escherichia & Klebsiella	30,747,106
	Pasteurellaceae/ Haemophilus	30,747,106
	Ruminococcaceae/Pseudoflavonifractor & Subdoligranulum & Faecalibacterium	30,747,106; 31,065,565; 29,335,554
	Veillonellaceae/ Veillonella & Dialister	30,232,462; 30,747,106
Proteobacteria	*Desulfovibrionaceae/ Bilophila & Desulfovibrio*	30,747,106
	Enterobacteriaceae/ Escherichia & Klebsiella	30,747,106
	Gamma-proteobacteria	30,232,462
	Pasteurellaceae/ Haemophilus	30,747,106
Verrucomicrobia	*Akkermansiaceae/ Akkermansia*	30,747,106

Footnote:

Curated list of taxa (phyla, family, and genera) derived from research studies that investigated (**a**) oral and (**b**) gut microbiome in healthy individuals. To identify these taxa, a conventional literature search was performed in PubMed, PubMed Central, Medline, and Google Scholar using the MeSH terms oral microbiome, gut microbiome, gastrointestinal microbiome, oral bacteria, flora, microbiome, microbiota, microflora, gut, oral, enteric, intestinal, and gastrointestinal. (**a**) Genera of the oral microbiota characterized by high transmission rates to the gut are indicated (**bold**). (**b**) Families and genera associated with temporal stability of the gut microbiome are indicated (**bold**)

producing bacteria diffuse from the lumen across the mucus layer contributing to further damage to ISCs (Fig. 1).

Objectives

Our first objective was to use conventional literature search methodology to examine associations of oral and gut microbiomes (fungi excluded) with aGVHD and other clinical outcomes in HSCT patients. In addition, clinical studies have described TLR4/LPS signaling pathways as being of interest in aGVHD outcome [20, 21]. Therefore, we sought to confirm and expand on previous findings using a computational approach. Accordingly, our second objective was to use data mining tools to expand the conventional literature search and to identify oral and gut lipopolysaccharide (LPS) producing bacterial taxa affected by drugs used for treatment of aGVHD, possibly impacting aGVHD drug targets.

Methods

A flow chart summarizing the methodology used is presented in Fig. 2.

Conventional review of the literature

Medline, PubMed, PubMed Central, and Google Scholar were searched for articles in English published between January 2008 and March 2020 using MeSH terms for GVHD, HSCT, oral microbiome, and gut microbiome. Duplicate articles were removed, and human studies retained. The top 100 articles obtained from each database, and their reference lists were examined for relevance. From the 400 articles, we selected 25 human studies using molecular techniques to determine gut and/or microbiome profiles, with/ without GVHD investigated as a clinical outcome. Experimental designs and outcomes of selected human studies are presented in Supplementary Table S1. Evaluation of bias was performed according to PRISMA guidelines (Supplementary Table S2).

Text mining

Of 400 articles, 90 were manually curated and selected as a positive "training" set ($n = 43$) or a negative training set ($n = 47$) based on relevance to aGVHD according to Biological Research Article Distiller v1.2 instructions [22]. BioReader extracts information from articles obtained via MeSH terms

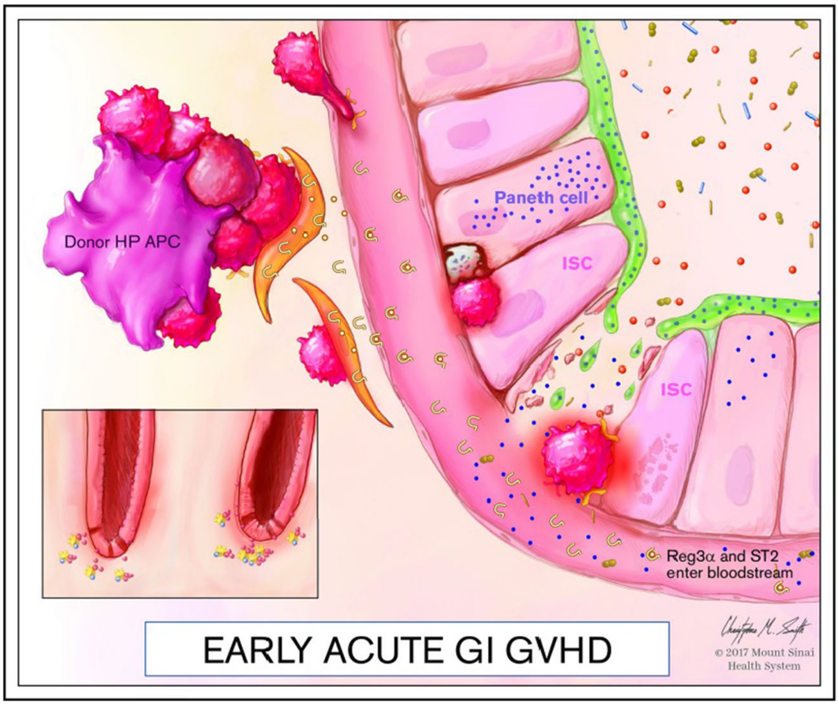

Fig. 1 Paneth cells in early GI aGVHD. Intestinal epithelium in the gastrointestinal (GI) tract is protected by interactions between microbial metabolites from bacteria in lumen that diffuse across the protective layer of mucus and interact with Paneth cells and intestinal stem cells (ISCs) in mucosal crypts. Paneth cells secrete antimicrobial proteins such as regenerating islet-derived protein 3 alpha (REG3α), able to kill gram-positive bacterial pathogens in the mucus layer maintaining cellular health of the GI epithelium. Bacteria are normally separated from the mucosal epithelium by a protective layer. Short-chain fatty acid butyrate (SCFA-butyrate) producing bacteria (i.e. *Lachnospiraceae, Peptostreptococcaceae,* and *Ruminococcaceae*) via fermentative metabolic pathways diffuse from the lumen across the mucus layer and fuel intestinal epithelial cells (IECs). SFCA-butyrate induces the differentiation of ISCs into mature innate lymphoid cells (ILCs) through histone acetylation, driving IL-22 secretion (Ferrara et al., 2018). REG3α works in conjunction with IL-22 and receptors for alarmin interleukin-33 such as ST2 to protect ISCs from the type of apoptotic signals which may occur during aGVHD episodes causing donor antigen presenting cells to attack healthy host tissues. Intestinal homeostasis is disrupted so that REG3α, donor derived ST2 leaks into the bloodstream (Ferrara et al., 2018). Levels of these two early biomarkers have been identified as predictive for patients who will develop severe aGVHD (Levine, 2014; Ferrara et al., 2018; Belizario et al., 2018; Haak et al., 2017). Additionally, Toll-like receptors (TLRs) on intestinal epithelial cells detect certain microbiota through microorganism associated molecular patterns (MAMPs) which trigger an inflammatory response that leads to the recruitment of T-cells to the GI tract and activation of the antigen presenting cells (APCs) that drive the adaptive immune response. (Kohler et al., 2019; Ferrara et al., 2018, Schwab, 2014; Socie, 2004; Giroux, 2011; Klambt, 2015, Hulsdunker, 2018; Holler et al., 2014). Permission to use image was given by Dr. James Ferrara, Ward-Coleman Chair in Cancer Medicine, Icahn School of Medicine at Mount Sinai, New York, NY

and computes statistics for the word counts used for classification. The positive training set included studies selected from the conventional search and studies identified with adjusted criteria to extract additional bacterial taxa possibly associated with aGVHD (Supplementary Table S3). Search terms in the training set articles were divided to create a document term matrix. Word counts were corrected using term frequency-inverse document frequency transformation, and Mann-Whitney U-test was used to reduce top terms differentiating the positive and negative training sets via BioReader v1.2 [23].

The "test" set for BioReader v1.2 was created through complementary PubMed search using the original search criteria described above. Using the 260 returned test articles, duplicates that appeared in both the returned PubMed articles and the manually curated positive or negative training sets were removed from the list of test articles. PubMed IDs were downloaded as a text file and uploaded into BioReader v1.2 for classification, ranking articles in order of relevance by using the maximum entropy algorithm based on top differentiating terms.

Of the 260 test articles, 103 were positively classified and manually filtered for relevance, resulting in the selection of 11 articles mentioning bacterial taxa identified as having a positive or negative correlation with aGVHD.

Microbiota LPS pathway and GVHD drug targets

Metabolic pathways were extracted from the online program Metabolic Pathways Database for Microbial taxonomic groups [24] to identify LPS producing bacterial taxa possibly involved in TLR4 activation or inhibition [18, 25]. ClinicalTrials.gov was searched to determine drugs completing trials for treatment of aGVHD and related

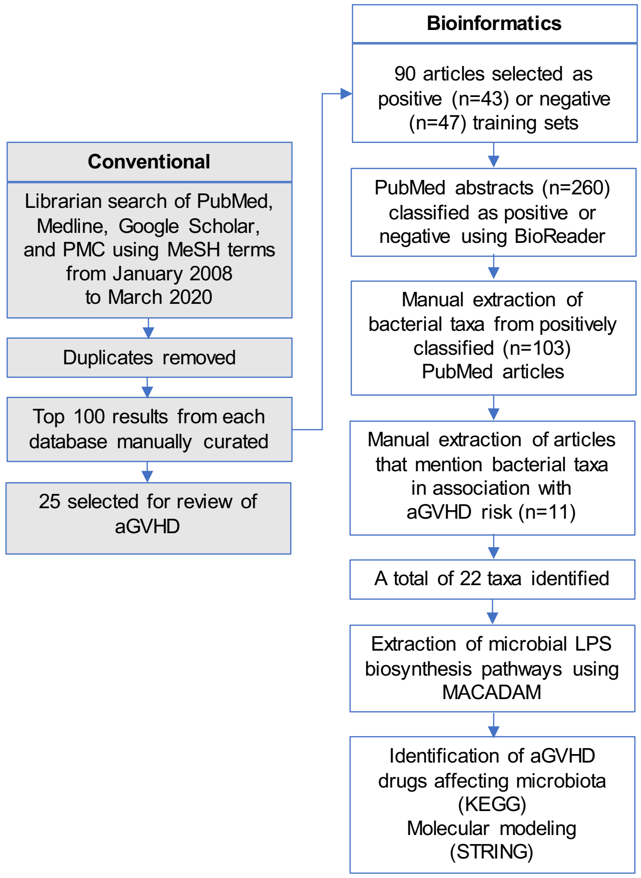

Fig. 2 Flow chart of conventional literature search complemented by bioinformatics tools. First, a conventional literature search was performed, independently, by a librarian at Atrium Health using Medical Subject Headings (MeSH) terms (gray flowchart, left) to extract information from (i) PubMed, (ii) U.S. National Library of Medicine (Medline), (iii) Google Scholar, and (iv) PubMed Central (PMC) databases, from January 2008 to March 2020. MeSH terms were as follows: hematopoietic stem cell transplant, HSCT, graft versus host disease, graft vs host disease, GVHD, mucositis, oral microbiome, gut microbiome, gastrointestinal microbiome, oral bacteria, flora, microbiome, microbiota, microflora, gut, oral, enteric, intestinal, and gastrointestinal. All duplicates and reprints of the articles were removed. A total of 400 abstracts (100 from each database) were manually curated and 25 articles were selected for review of acute graft versus host disease (aGVHD) based on relevance. To validate findings, BioReader online text mining tool (white flowchart, middle) was used based on an input of 90 articles curated from conventional methods and sorted into a positive test group (more relevant, $n = 43$) and negative test group (less relevant, $n = 47$). Then, a PubMed search was completed with BioReader using the original MeSH terms from the initial conventional literature search. All returned abstracts ($n = 260$), excluding duplicates and reprints, were processed in BioReader program for classification resulting in 103 positively classified articles. Manual extraction of articles mentioning bacteria in association with the risk of aGVHD ($n = 11$) was completed. A total of 22 bacterial species/genera identified across all articles were entered into the **M**et**A**boli**C** p**A**thways **DA**tabase for **M**icrobial taxonomic groups (MACADAM) online program and molecular pathways were returned. Molecular pathways were further analyzed to identify genes involved in lipopolysaccharide (LPS) biosynthesis. Identification of drugs used for treatment of aGVHD affecting microbiota was performed using **K**yoto **E**ncylopedia of **G**enes and **G**enomes (KEGG) database. STRING program was used to generate molecular interaction networks of the aGVHD drug targets identified

symptoms. Drug and compound IDs were used to identify aGVHD drug targets and microbial taxa affected by aGVHD drugs, using Kyoto Encyclopedia of Genes and Genomes (KEGG) database [26]. In addition, STRINGv11.0 program [27] was used to generate molecular interaction networks related to candidate aGVHD drug targets.

Results

Association of oral microbiome with non-GVHD outcomes in HSCT patients

Studies have reported non-GVHD outcomes associated with oral and gut microbiome profiles in HSCT patients. In a study by Galloway-Peña et al., oral samples and gut samples were prospectively collected in HSCT patients diagnosed with acute myeloid leukemia ($N = 59$), and the microbiomes were analyzed for associations with morbidity/mortality. However, associations with aGVHD were not reported [28]. Adverse outcomes were more closely associated with patients who displayed greater fluctuations in their microbiomes over time.

In a smaller microbiome study on oral samples from HSCT patients ($n = 11$), in which aGVHD was not considered as an outcome, respiratory complications up to 100 days post-transplant were attributed to mean abundance differences of several bacterial species [29]. Lower mean abundance levels of *Streptococcus* species (including *S. oralis* and *S. infantis*) and *Rothia* species (including *R. dentocariosa* and *R. mucilaginosa*) and higher mean abundance levels of *Veillonella* species (*V. parvula* and *V. atypica*), *Streptococcus* species (*S. parasanguinis* and *S. anginosus*), *Peptostreptococcus micros*, and *Gemella haemolysans* were observed in the group with respiratory complications. However, no significant temporal changes in the oral microbiome were found. The abovementioned studies and nine other studies investigating associations of the oral and/or gut microbiome profiles with non-GVHD outcomes are presented in Supplementary Table S1a, with the majority analyzing both gut and oral samples.

Association of oral (and gut) microbiome with aGVHD in HSCT patients

The standalone study by Golob et al. identified oral bacterial taxa in the stool positively correlated with GVHD [30]. Approximately 100 days post-HSCT, the presence of oral *Actinobacteria* and oral *Firmicutes* (Table 2a), and some gut bacterial genera in stool, were positively correlated with aGVHD; gut *Lachnospiraceae* were negatively correlated (Table 2b). Members of the *Bacteroides* genus

(*B. thetaiotaomicron, B. ovatus, and B. caccae*) were negatively correlated with severe aGVHD, while others (*B. dorei*) were positively correlated. Gram-positive bacterial species, presumed of oral origin, such as *Rothia mucilaginosa, Solobacterium moorei, and Veillonella parvula* (Table 2a), were positively correlated with aGVHD, whereas the gut *Lachnospiraceae* species *Blautia luti* and *Butyricicoccus spp.* of the *Clostridiaceae* family were negatively correlated (Table 2b).

A relative abundance (RA) gradient of stool bacterial species was most predictive of severe acute GVHD, with a receiver operator characteristic area under the curve of 0.83. Moreover, the stool microbiota around the time of neutrophil recovery post-HSCT was predictive of severe aGVHD. The experimental design of this study is presented in Supplementary Table S1b.

Furthermore, BioReader program positively classified a study that sampled oral specimens of pediatric patients after HSCT [31]. Both patients who developed aGVHD and those who did not exhibited a parallel trend in Shannon diversity index (SDI) changes in oral samples 24 days post-HSCT.

Association of gut microbiome with aGVHD in HSCT patients

We identified several studies investigating gut microbiome associations with aGVHD by using conventional literature search [32–37]. In a study by Holler et al., serial stool specimens were collected from 31 HSCT patients to analyze the gut microbiome during treatment [32]. In post-transplant HSCT patients, the authors determined a microbiome shift associated with gut aGVHD, involving an increase in *Enterococcus species*, especially in patients receiving antibiotics for neutropenic infections [32]. An increase in opportunistic bacteria and a decrease in overall bacterial load was associated with aGVHD.

In another study, changes in abundances of bacterial species in gut including taxa presumed of oral origin were shown to be associated with morbidity and mortality in allogenic HSCT patients [33]. Serial stool samples were collected from 44 patients before, during, and up to 100 days post-transplant. A lower SDI at the time of engraftment (14 to 28 days post-conditioning) was associated with increased incidence of gut aGVHD and transplant-related deaths as has also been shown

Table 2 Oral and gut bacterial taxa possibly associated with aGVHD

a. Oral microbiome		
Phylum	Family/Genus	PubMed ID
Actinobacteria	*Micrococcaceae/ Rothia- R. mucilaginosa* (+)	29,020,185
Firmicutes	*Erysipelotrichidae/ Solobacterium- S. moorei* (+)	29,020,185
	Veillonellaceae/ Veillonella- V. parvula (+)	25,977,230; 29,020,185
b. Gut microbiome		
Actinobacteria	*Coriobacteriaceae* (−)	31,065,565
	Micrococcaceae/ Rothia – R. mucilaginosa (+)	29,020,185
Bacteroidetes	* *Bacteroidaceae/ Bacteroides- B. dorei* (+) & *B. thetaiotamicron* (−) & *B. ovatus* (−) & *B. caccae* (−)	29,020,185
Firmicutes	*Clostridiaceae/ Butyricicoccus (−)*	29,020,185
	Enterococcaceae/ Enterococcus (+)	24,492,144; 25,977,230
	**Erysipelotrichaceae/ Erysipelatoclostridium* (−) & *Solobacterium -S. moorei* (+)	29,020,185
	Lachnospiraceae/ Blautia luti (−) & *Lachnoclostridium (−)*	30,845,942; 29,020,185; 22,547,653; 25,977,230; 29,226,172
	Peptostreptococcaceae (−)	31,299,215
	Ruminococcaceae (−)	25,977,230; 28,967,895
	**Veillonellaceae/ Veillonella – V. parvula* (+)	25,977,230; 29,020,185
Fusobacteria	*Fusobacteriaceae/ Fusobacterium (+)*	30,845,942
Proteobacteria	*Enterobacteriaceae* (+)	31,299,215; 25,977,230; 29,226,172

Footnote:

Phyla, family, and genera curated from text mining of publications using conventional methods that investigated the (a) oral and (b) gut microbiome in individuals with aGVHD. All oral bacteria were identified from samples in the stool. *Genera with species identified as being associated with aGVHD. aGVHD positively (+) and negatively (−) correlated bacterial taxa are shown. aGVHD is acute graft versus host disease in other studies [38, 39]. However, the incidence of the family *Coriobacteriaceae* (phylum *Actinobacteria*) was negatively correlated with gut aGVHD [33] (Table 2b).

A multicenter microbiome study analyzed stool samples from 36 pediatric patients before HSCT, during engraftment, and > 30 days post-HSCT [34]. A pre-HSCT dysbiotic gut microbiome profile was predictive of patients who developed gut aGVHD. Reduced diversity, lower *Blautia* content (phylum *Firmicutes*, class *Clostridia*, family *Lachnospiraceae*), and increased abundance of *Fusobacterium* (phylum *Fusobacteria*) post-HSCT characterized such predisposition (Table 2b). The first samples taken after engraftment (12 to 28 days post-HSCT) showed a deep rearrangement of the gut microbiome structure occurring in all patients but recovered to a more eubiotic balance 30 days post-HSCT [34].

Six other studies employed various 16S rRNA gene next-generation sequencing to analyze the association of the gut microbiome with GVHD and/or morbidity and mortality in HSCT patients [16, 35, 39–41]. These studies agreed that either just before or during HSCT, a dysbiosis in the gut microbiota occurred, allowing changes in abundance of certain microbiota that were associated with increased risk for gut aGVHD or higher morbidity in HSCT patients. These changes included increased levels of the family *Enterobacteriaceae* (phylum *Proteobacteria*, class *Gammaproteobacteria*) and decreased levels of the health-associated families *Lachnospiraceae and Ruminococcaceae* (phylum *Firmicutes*, class *Clostridia, genus Blautia*) [35, 39–41] (Table 1b and 2b).

In addition, our BioReader analysis classified seven studies investigating the association of gut microbiome and aGVHD or chronic GVHD [36, 42–45]. In pediatric HSCT patients, aGVHD was found to be associated with a RA increase for *Enterococcus* and Clostridiales and a RA decrease in *Faecalibacterium* and *Ruminococcus* in fecal samples. The authors found a RA decrease of *Firmicutes* up to 35 days post-HSCT, which then was restored from days 35 to 65 to a level that was higher than at baseline sampling. In cases of aGVHD, *Bacteroidetes* were found to have lower abundances across all time points than in non-GVHD diagnosed patients. After HSCT, there was an increase in *Firmicutes*, *Proteobacteria*, and *Spirochaetes*. Prior to HSCT, pediatric patients who developed aGVHD had lower diversity and richness, lower *Bacteroides*, and *Parabacteroides* [42].

Another positively classified study described that a higher RA of *Eubacterium limosum* in the gut was associated with a decreased risk of chronic GVHD relapse and a 2-year incidence relapse for HSCT patients with higher *E. limosum* RA [44]. Multiple positively classified studies confirmed findings that lower gut microbial diversity of the recipient was associated with a higher incidence of aGVHD [36, 42, 43, 45]. One study reported that higher gut microbiome diversity in the donor was associated with less risk of developing aGVHD post-HSCT [41]. The experimental design of some of the abovementioned aGVHD studies is presented in Supplementary Table S1b.

Diversity and variability in oral vs. gut microbiome in HSCT patients

In the study by Galloway-Peña et al., discussed above (non-GVHD outcome in HSCT patients), the authors found no major microbiome structural composition differences between oral and stool samples, regarding either the SDI coefficient of variation (i.e., CV values) or other metrics of alpha-diversity (Chao-1 diversity index and Simpson's diversity index) [28]. In fact, the SDI CV of oral and stool samples from the same patients were moderately correlated for temporal variability.

The authors postulated that this relationship between the two sites may reveal that factors in these AML patients influence the temporal variability of microbial diversity at the oral and gut sites concurrently. When the RA of bacterial populations was examined, certain stabilizing and destabilizing taxa were identified, some common to both oral and stool samples. In oral samples, temporal instability was influenced by increasing RA of pathogenic-associated genera such as *Staphylococcus*, *Streptococcus* and *Stenotrophomonas*. Similarly, in stool samples, instability of community structure was statistically positively correlated with the abundance of *Staphylococcus* and *Streptococcus*. The same genera were also correlated with temporal instability of microbial α-diversity (SDI CV) in oral and stool samples. Instead, the abundance of nonpathogenic organisms in the stool, such as *Akkermansia, Subdiloganulum, and Pseudobutyrivibrio* was associated with increased temporal stability of both the microbiome community structure and diversity.

These data imply that patients with a high RA of commensal organisms, like *Akkermansia*, can maintain a more stable microbiome during their hospitalization whereas patients with higher RA of pathogenic-associated bacteria, like *Staphylococcus and Streptococcus*, cannot maintain temporal stability of both the oral and gastrointestinal microbiome.

Microbial LPS biosynthesis, TLR4-related pathways, and aGVHD drugs

TLR4 is a specific toll-like receptor for LPS that has been a target of interest for treatment of aGVHD [20]. Indeed, one study that observed TLR4 single nucleotide polymorphisms found that TLR4 (rs17582214 and rs4837656, located in promoter region) was associated with a higher incidence of aGVHD, possibly due to a toxic gain in function [21]. However, issues with targeting TLR4 include altering production of interferon-gamma, IL-1B, IL-6, IL-12, and TNF-α, which can reduce the effectiveness of HSCT [20].

No clinical trials were found that target TLR4 in the treatment of GVHD or its symptoms. Nevertheless, one anti-TLR4 drug, NI-0101, was recently investigated for its effectiveness in reducing the symptoms of rheumatoid arthritis after TLR4 inhibition was confirmed by in vivo LPS-induced cytokine release [46]. Phase II trials showed that 52.5% of rheumatoid arthritis patients who received the drug had adverse events during treatment [46].

Meanwhile, a pilot study by our team, investigating the association the oral microbiome with aGVHD, showed the presence of a higher average RA of *Capnocytophaga spp.* in oral samples of HSCT patients who did not develop aGVHD (Supplemental File Appendix) [47]. LPS produced by the periodontopathic bacterial species *C. ochracea* has been shown to inhibit TLR4 [48]. While oral to gut transfer rate of *Capnocytophaga spp.* is unknown and confirmatory investigation is needed, this result suggests the possibility of a mechanism which might be targeted to mitigate aGVHD without causing significant adverse events in HSCT patients.

In addition, our analysis by MACADAM Explore determined that all phyla identified from BioReader positively classified articles had metabolic pathways involving LPS biosynthesis. *Proteobacteria* and *Firmicutes* phyla had one or more article that associated an increase in RA to aGVHD development. MACADAM Explore also identified the order *Clostridiales* as possessing LPS metabolic pathways, but only one of the three articles associated an increase in RA with GVHD (Table 3). Furthermore, drugs being investigated for the treatment/prevention of aGVHD may affect oral and gut microbiota (Table 4), thereby influencing the impact of LPS. Target genes of these drugs were processed with STRING online program to identify the nearest neighbor molecular network with high (Fig. 3a) and low (Fig. 3b) level of confidence. The network combining GVHD drug targets shows that molecular pathways surrounding TLR4 signaling may be targeted in GVHD drug development.

Discussion

Our study showed that both oral and gut microbiome could play a significant role in the development of aGVHD, if an imbalance of potentially pathogenic species versus commensals occurs (dysbiosis) pre- or post-HSCT. This is the first comprehensive systematic review combining conventional literature search and bioinformatics tools to refine the analysis. We were able to confirm the implication of bacterial taxa in aGVHD and expand findings from conventional methods. We used MACADAM Explore to find microbiota taxa and associated LPS pathways that could represent biomarkers or drug targets to prevent aGVHD in the GI tract. The oral microbiome is the second most diverse site in the human body, yet little research has been done to investigate microbial changes in the oral microbiome associated with aGVHD.

Species in the genus *Streptococcus* have been associated with gut dysbiosis in healthy patients while *Akkermansia*, *Subdiloogranulum*, and *Pseudobutyrivibrio* have been shown to increase temporal stability in cancer patients [28]. Dysbiosis associated with risk for gut aGVHD has been characterized by increased levels in the family *Enterobacteriaceae* and decreased levels in the families *Lachnospiraceae* and *Ruminococcaceae* [35, 39–41]. Holler et al. reported that the average RA of enterococci was higher in aGVHD vs. non-aGVHD patients, whereas Galloway-Peña et al. found that *Coriobacteriaceae* was negatively correlated with aGVHD [32, 33]. BioReader analysis showed that bacteria exhibited increased RA and potentially activated LPS pathways during aGVHD, whereas the majority of those with decreased RA did not have associated LPS pathways (Table 4). Further investigation of LPS producing bacteria could identify their function in exacerbation (or prevention) of aGVHD.

In pediatric HSCT patients, higher levels of the genera *Enterococcus* and *Clostridiales* and lower levels of *Faecalibacterium* and *Ruminococcus* were reported. Patients that had aGVHD were associated with *Bacteroides* levels being lower at all time points of the study with *Parabacteroides* being lower prior to HSCT. Post-HSCT patients that developed aGVHD were associated with higher levels of *Firmicutes*, *Proteobacteria*, and *Spirochaetes* [42]. Another study reported that *Eubacterium limosum* in the gut was associated with an overall decreased risk of relapse of disease [44].

Current treatments for aGVHD include antibiotics or immunosuppressants that can further agitate the symptoms associated with aGVHD. Furthermore, clinical trials have shown that fecal microbiota transplantation (FMT) may reduce aGVHD symptoms by improving GI microbiome diversity. In a pilot study, eight FMT treated GI aGVHD patients had increased beneficial bacteria and a decrease in clinical symptoms [5]. Another pilot study found increased gut microbiome diversity in HSCT transplant recipients when FMT was administered early after allogenic HSCT, suggesting partial eubiosis restoration [49]. FMT has been shown to maintain gut homeostasis and reduce symptoms of aGVHD in the gut [5], but more research is warranted to determine the effects over time.

One in three oral species has been reported as being transferable to the gut with even higher levels of transfer in diseases like colorectal cancer and rheumatoid arthritis. This is especially true for opportunistic bacteria [14]. In HSCT, patient's temporal variability in the gut and oral microbiome was associated with a higher risk of infection. *Staphylococcus*, *Streptococcus*, and *Stenotrophomonas* have been associated with temporal instability in oral samples of cancer patients [28], whereas *P. gingivalis* and *F. nucleatum* were able to alter the immune-inflammatory state of the host and its biofilm

Table 3. BioReader positively classified articles associated with aGVHD and biosynthesis of lipopolysaccharide (LPS)

Bacterial taxa[a]	Taxonomic level[b]	LPS Pathways[c]	BioReader GVHD association[d]	PMID[e]
Bacteroidetes	phylum	Lipid IVA biosynthesis; Kdo transfer to lipid IVA I	Decrease Decrease	25,893,458 28,733,895
Clostridiales	order	Lipid IVA biosynthesis	Increase Decrease Decrease	31,042,160 25,893,458 27,856,471
Enterobacter	genus	polymyxin resistance; enterobacterial common antigen biosynthesis; Kdo transfer to lipid IVA I; lipid IVA biosynthesis	Increase	27,856,471
Enterobacteriaceae	family	*Escherichia coli* serotype O9a O-antigen biosynthesis; polymyxin resistance; enterobacterial common antigen biosynthesis; Kdo transfer to lipid IVA I; lipid IVA biosynthesis; *E. coli* serotype O86 O-antigen biosynthesis; D-galactosamine biosynthesis	Increase Increase Increase	29,740,427 31,299,215 31,519,210
Escherichia	genus	lipid IVA biosynthesis; Escherichia coli serotype O86 O-antigen biosynthesis; Kdo transfer to lipid IVA I; D-galactosamine biosynthesis; *E. coli* serotype O9a O-antigen biosynthesis; enterobacterial common antigen biosynthesis; polymyxin resistance	Increase	27,856,471
Eubacterium	genus	ND	Decrease Decrease	28,296,584
Firmicutes	phylum	lipid IVA biosynthesis; Kdo transfer to lipid IVA II	Increase Increase Decrease	25,893,458 28,733,895 31,042,160
Klebsiella	genus	lipid IVA biosynthesis; enterobacterial common antigen biosynthesis; polymyxin resistance; Kdo transfer to lipid IVA I; *E. coli* serotype O9a O-antigen biosynthesis	Increase	27,856,471
Lachnospiraceae	family	ND	Decrease Increase	29,740,427 31,299,215
Lactobacillus	genus	ND	Increase Increase Increase	27,856,471 31,519,210 31,042,160
Parabacteroides	genus	Lipid IVA biosynthesis	Decrease	31,042,160
Peptostreptococcaceae	family	Lipid IVA biosynthesis	Increase	31,299,215
Proteobacteria	phylum	Kdo transfer to lipid IVA II; Kdo8N transfer to lipid IVA; D-galactosamine biosynthesis; *E. coli* serotype O9a O-antigen biosynthesis; polymyxin resistance; *E. coli* serotype O86 O-antigen biosynthesis; Kdo transfer to lipid IVA I; lipid IVA biosynthesis; enterobacterial common antigen biosynthesis	Increase	25,893,458
Ruminococcaceae	family	ND	Decrease Decrease	29,740,427 31,519,210
Spirocheta	genus	ND	Increase	25,893,458
Streptococcus	genus	lipid IVA biosynthesis	Increase	27,856,471
Bifidobacterium	genus	ND	Decrease	31,566,729

Footnote:

Two of four families identified in one or more articles that described an increase in relative abundance (RA) associated with aGVHD as having discovered LPS metabolic pathways via MACADAM Explore. The other two families had one or more articles that found a decrease in RA to be associated with aGVHD and no discovered LPS pathways in the MACADAM Explore database

Four of the 12 genera had increased RA in aGVHD patients and possibly activated LPS metabolic pathways. Three of the 12 had decreased RA and no identified LPS metabolic pathways. The other five either had no known LPS metabolic pathways and articles described finding an increase in RA associated with aGVHD ($n = 4$), or LPS metabolic pathways identified by MACADAM Explore and a decrease in RA associated with aGVHD ($n = 1$). The two species (i.e., *Eubacterium limosum*, and *Lactobacillus rhamnosus)* had lower RA in aGVHD patients, although no associated LPS metabolic pathways have been discovered

[a] Bacterial taxa from [e] BioReader positively classified articles and [b] their taxonomic levels showing [c] lipopolysaccharide biosynthesis pathways identified using MACADAM Explore online tool which curated all possible pathways for the level of taxonomy and then LPS biosynthesis pathways were searched from each returned dataset. [d] Changes in RA of bacterial taxa associated with aGVHD from [e] BioReader articles. Articles were determined to be relevant using the max entropy algorithm with a "positive" group of articles having a Precision of 0.876, Recall of 0.93, and an F score of 0.884. Not shown are *Blautia*, *Eubacterium limosum*, and *Lactobacillus rhamnosus,* all of which had higher levels of taxonomy already shown to contain no known LPS biosynthesis pathways. ND is Not Determined

Table 4 aGVHD preclinical or clinical trial drugs affecting bacteria

Drug name[a]	Drug description[b]	Bacterial taxa affected[c]	KEGG Drug ID[d]	KEGG Target[e]	PMID[f]
Etanercept; Enbrel	Dimeric fusion protein	*Cyanobacteria* increase in rheumatoid arthritis patients	D00742 C07897	TNFA TNFSF2 TNFB TNFSF1	30,261,687 29,073,260 28,405,499
Tacrolimus; Tacrolimus hydrate; Prograf; Protopic	Immuno-suppressant	*Clostridium, Ruminococcus, Rikenella, Ruminococcaceae, Oscillospira* decrease in mice; *Faecalibacterium* increase in kidney transplant patients	D00107	PPP3C PPP3-R CHP	29,316,256 25,815,766 31,595,800
Methotrexate; Otrexup; Xatemp	Enzyme inhibitor	*Ruminococcaceae, Bacteroidales uniforms, Bacteroidales fragilis* decrease in mice; *Lachnospiraceae and Bacteroidales thetaiotamicron* increase in mice; Decrease in *Enterobacteriales* patients with rheumatoid arthritis	D00142 C01937	DHFR	30,049,384 30,261,687 28,784,598 24,982,504 30,824,040
Rapamycin; Sirolimus; Rapamune	Macrolide compound from *Streptomyces nygroscopic*	*Lactobacillus intestinalis*, *Rikenellaceae OTU 1056* increase; *Acidobacteriaceae* and *Rikenellaceae OTU 1068* increase in immune-deficient mice	D00753 C07909	MTOR	26,315,673 19,597,415 29,445,370 24,980,254
Ustekinumab; Stelara	Human immunoglobulin G1 kappa monoclonal antibody	*Faecalibacterium, Blautia, Clostridium XIVa, Ruminococcaceae,* and *Roseburia* increase in Random Forest model of Crohn's Disease patient outcomes. *Coprococcus* increase in long term use Psoriasis patients	D09214	IL-12 IL-23A	29,535,202 31,549,347 29,242,294 21,874,061
Mycophenolate Mofetil; Cellcept	Reverse inhibitor	Decrease of *Bacteroidetes*, *Verrucomicrobia*, *Akkermansia*, *Parabacteroides*, and *Clostridium* in mice; Increase in *Proteobacteria (Escherichia/Shigella)*, and *Deferribacteres* in mice	D00752	IMPDH	30,173,823 30,824,040 22,592,321 25,061,777 24,418,750
Atorvastatin Calcium; Lipitor	Statin	Decrease of *Bacteroides, Butyricimonas*, and *Mucispirillum* in mice with hyperglycemia and hyperlipidemia. *Faecalibacterium prausnitzii, Akkermansia muciniphila, Oscillospira* increase; and decrease in *Desulfovibrio sp., Bilophila wadsworthia,* and *Bifidobacterium bifidum* in hypercholesterolemic treated patients	D02258	HMGCR	31,551,944 29,432,061 24,958,183 28,412,518 26,256,940
Cyclophosphamide; Cytoxan; Neosar	Alkylating agent	In Mice, *Clostridia XIVa, Roseburia, Lachnospiraceae unclassified, coprococcus, Lactobacilli and Enterococci* decreased. *Bacteroidetes, Prevotellaceae, S24–7*, *Alcaligenaceae, Rodospirillaceae, Strptococcaceae* decrease in mice and an increase in *Firmicutes, Bacilli, Clostridia, Coriobacteria, and Mollicutes*	D00287 C06933	DNA	24,264,990 29,535,202 25,240,817 30,194,866 28,395,546
Cyclosporine; ciclosporin; Gengraf; Neoral; Restasis; Sanimmune	Cyclosporin	No bacteria affected over time in the oral microbiome of Severe Aplastic Anemia. *Clostridium cluster I, Clostridium cluster XIV, Enterobacteriaceae* decrease and an increase of *Faecalibacterium prausnitzii* in the GI tract of Orthotropic liver transplant patients	D00184 C05086	PPP3C PPP3-R CHP	30,919,073 31,020,543 28,395,546 30,289,329 26,808,568
NI-0101	Anti-toll-like receptor 4	ND	ND	TLR-4	31,892,533 27,706,798

Footnote:

Description of [a] clinical trial drug identifying name/s being investigated for the treatment of aGVHD, their [b] function in human subjects and [c] bacterial taxa of oral cavity or gut affected by the drug in mice or humans. [d] Kyoto Encyclopedia of Genes and Genomes (KEGG) database drug and compound IDs were used to identify [e] drug targets. [f] Pubmed ID. ND is not determined

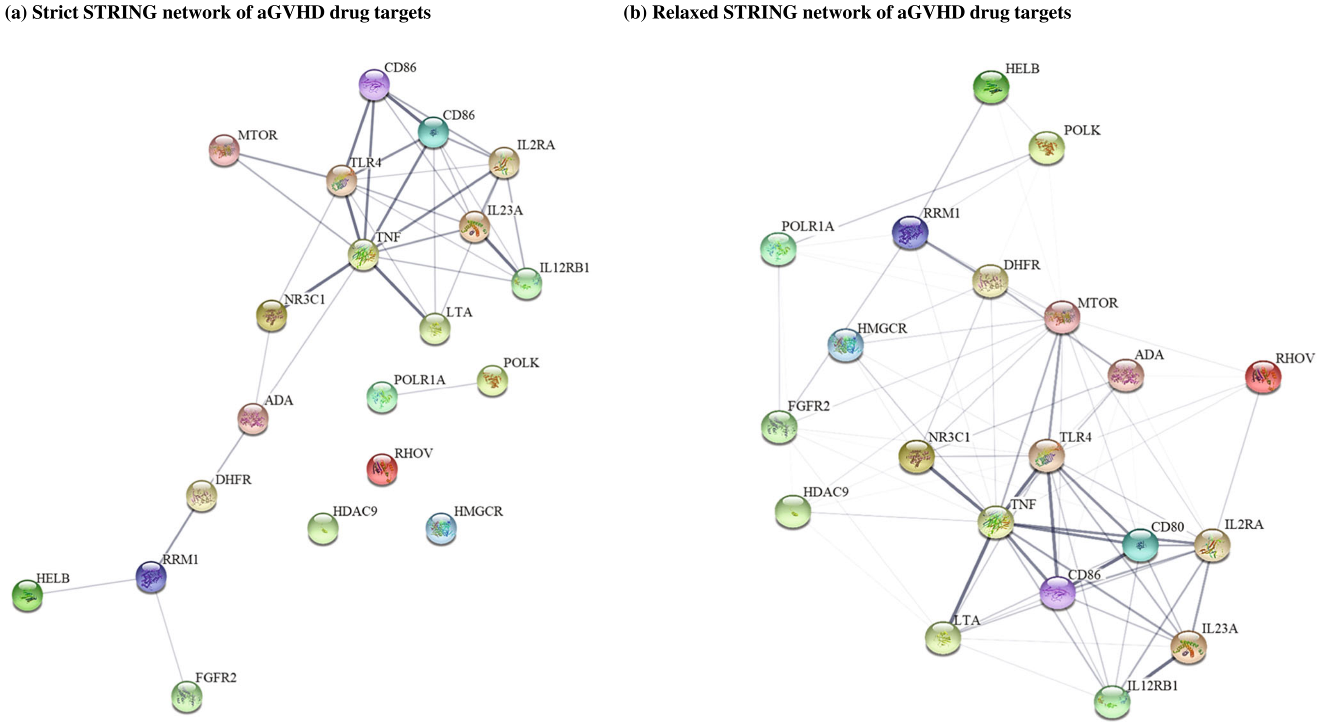

Fig. 3 **STRING: functional protein association networks of aGVHD drug targets 3a. Strict STRING network of aGVHD drug targets 3b. Relaxed STRING network of aGVHD drug targets.** Protein-Protein interactions of aGVHD clinical drug trial targets with network edges showing confidence of interactions using STRING's "active interaction sources" at a minimum interaction score of **a.** 0.40 and **b.** 0.150, and no maximum number of interactors. Confidence of interactions are portrayed by thickness of line. Drug targets not shown: PPP3C, PPP3R, and IMPDH

[15]. Therefore, it will be important to understand the temporal changes and differences in mean abundance of certain bacterial taxa in larger controlled studies to determine associations with aGVHD.

Finally, even though TLR4 has been investigated as a drug target in other inflammatory diseases like rheumatoid arthritis [46], more preclinical trials with aGVHD drug candidates are warranted along with findings new means to mitigate the side effects observed in past clinical trials. The side effects observed in one study suggest that earlier targeting of LPS biosynthesis might reduce aGVHD symptoms [46]. In the presence of bacteria-producing LPS, the differentiation of TLR4 may be induced. Thus, early targeting of gut bacterial species having lipid IVA biosynthesis pathways may reduce inflammation earlier without disrupting cytokine production. This early action could also promote a graft versus leukemia effect beneficial to cancer patients [20].

Limitations

Conventional methods of systematic reviews require manual curation of information in which some key elements may be overlooked. Meanwhile, text mining tools rarely perform full article searches including figures and tables, leaving the investigator to verify abstract content. In addition, microbial pathways associated with aGVHD were not found in human disease databases, hindering our ability to take a closer look at bacterial taxa already identified as potential biomarkers for aGVHD. Furthermore, current clinical drug trial information for aGVHD was scarce.

Conclusions

Although aGVHD can affect the oral cavity, little has been explored about the changing oral microbial levels and interactions with gut microbiome in patients with aGVHD. Dysbiosis of the gut microbiome is clearly related to HSCT. The composition of the host-associated microbial communities is severely perturbed during critical illness. Reduced microbial diversity reflects high illness severity and is associated with mortality. The association of oral bacterial species with aGVHD and their interactions with species of the GI tract remain a matter of active research requiring the use of multi-omics experimental approaches and bioinformatics tools.

Authors' contributions Dr. Farah Bahrani Mougeot and Dr. Jean-Luc Mougeot conceived the study. Darla Morton conducted the conventional literature search and compiling of summaries. Micaela Beckman conducted verifications of the conventional literature search and complemented the study with the bioinformatics approach. All authors had significant contributions in writing and revisions of main manuscript, tables, and figures.

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